

Ab Initio Whole Cell Kinetic Model of *Bifidobacterium breve* DSM 20213 (bbrdPS26)

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Citation: Subramanian P, et al. Ab Initio Whole Cell Kinetic Model of *Bifidobacterium breve* DSM 20213 (bbrdPS26). J Clin Immunol Microbiol. 2026;7(3):1-6.

<https://doi.org/10.46889/JCIM.2026>

Received Date: 07-09-2026

Accepted Date: 21-09-2026

Published Date: 28-09-2026



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Abstract

Bifidobacterium breve is an important constituent of healthy gut microbiota and has been shown to prevent paediatric intestinal diseases as well as Alzheimer's pathologies in the elderly. This makes it a potential metabolic engineering candidate for enhanced functional properties. Whole-cell Kinetic Models (KMs) are useful computational tools to guide engineering efforts and there has been no whole-cell KM of *B. breve* to date. This study aims to address this gap by developing a whole-cell KM for *B. breve*. The resulting KM; denoted as bbrdPS26, is based on the annotated genome of *B. breve* DSM 20213; consists of 580 metabolites, 329 enzymes with corresponding transcriptions and translations and 453 enzymatic reactions. Thus, presenting a baseline whole-cell KM of *B. breve* for future development.

Keywords: Whole-Cell Kinetic Models; *Bifidobacterium breve*; Alzheimer's

Introduction

Bifidobacterium breve is a Gram-positive, a non-motile and an anaerobic bacterium that belongs to the phylum Actinobacteria. It is an important gut bacterium to support the normal immune function [1-3]. It has also been shown to be beneficial in preventing and treating paediatric intestinal diseases such as acute infectious diarrhoea [4,5]. Furthermore, *B. breve* may also improve Alzheimer's pathologies in the elderly [6-8]. Hence, there is also an interest in engineering *B. breve* for enhanced or tailored functional properties [9,10].

Before experimental work begins, metabolic engineering often turns to mathematical modelling to explore feasible intervention strategies [11,12]. Two frameworks dominate: GSMs and KMs [13,14]. GSMs, known for their constraint-based formulation, mainly

predict fluxes. KMs offer broader predictive capacity by modelling yields as well as rates [15]. They also make *in-silico* gene knock-ins more straightforward to implement [16]. Because of these advantages, KMs provide a more practical platform for evaluating engineering scenarios. In recent years, this has led to growing advocacy for constructing more robust kinetic models that can better support early-stage decision-making [17,18].

Although a whole-cell KM of *Bifidobacterium longum* JCM1217 has recently been published; there has been no whole-cell KM of *B. breve* to date [19]. Hence, this study aims to construct a KM of *B. breve* DSM 20213 using ab initio approach by identifying enzymes from its published genome and identifying the corresponding reaction from KEGG [20,21]. The result is a whole cell KM of *B. breve* DSM 20213, named as bbrdPS26 using the nomenclature proposed by Cho and Ling, which consists of 580

metabolites, 329 enzymes with corresponding transcriptions and translations and 453 enzymatic reactions; and has been deposited into KinModRe [22,23].

Materials and Methods

Identification of Reactome

The reference genome of *Bifidobacterium breve* DSM 20213 (= JCM 1192) was obtained from the NCBI GenBank database where it is archived under the assembly accession GCF_001025175.1 (accession number NZ_AP012324.1) [20]. The genome assembly was used as the primary reference to identify enzymatic genes focusing on complete Enzyme Commission (EC) number was mapped to its corresponding reaction identifier using the KEGG LIGAND database for enzyme nomenclature. For example, EC 1.1.1.23 catalyses reactions 401158, R01163 and R03012, wherein substrates and products of each reaction can be identified.

Model Development

The model was developed using the ODE formats described in Sim, et al., [24]. By integrating BioNumbers figures, we estimate that *Escherichia coli* synthesizes about 5600 kDa of RNA per second from about 3000 RNA polymerase molecules (BioNumbers 106199) of which, about 25% are active (BioNumbers 111676) [25,26]. Given elongation rate of 22 nucleotide per second (BioNumbers 104109) and 339.5 Da per nucleotide, this output converts to 9.3e-18 grams of RNA per second. Relative to a cell volume of 7e-16 litres and 4225 coding genes (BioNumbers 105443), transcription yields 2.92 micromolar per gene per second. With an average lifespan of 107.56 seconds (BioNumbers 107666), the mRNA decay rate is 0.0093 per second; thus, giving $d[\text{mRNA}]/dt = 0.00292 - 0.0093[\text{mRNA}]$ [27-30]. Protein synthesis at 0.278 peptides per second per mRNA (BioNumbers 106382) together with a protein decay rate of 2.78e-6 per second (BioNumbers 109924) produces $d[\text{peptide}]/dt = 0.278[\text{mRNA}] - 0.00000278[\text{peptide}]$ [31,32]. The complete kinetic system was encoded as ODEs using median parameters (k_{cat} of 13.7 per second and K_m of 1 mM) following AdvanceSyn model format [33-36].

Model Simulation

The constructed model was tested for simulatability using AdvanceSyn Toolkit [36]. Initial concentrations of all mRNA species and enzymes were set to 0 mM. Most metabolites concentrations were initialised at 1 mM. However, a subset of metabolites was assigned an initial concentration of 1000 mM to ensure their availability through the simulation: (I) C00001 (Water), (II) C00002 (ATP), (III) C00003 (NAD⁺), (IV) C00004 (NADH), (V) C00005 (NADPH), (VI) C00006 (NADP⁺), (VII) C00007 (Oxygen), (VIII) C00011 (Carbon Dioxide), (IX) C00014 (Ammonia), (X) C00025 (L-Glutamate), (XI) C00031 (D-Glucose), (XII) C00037 (Glycine), (XIII) C00041 (L-Alanine), (XIV) C00047 (L-Lysine), (XV) C00049 (L-Aspartate), (XVI) C00064 (L-Glutamine), (XVII) C00065 (L-Serine), (XVIII) C00073 (L-Methionine), (XIX) C00097 (L-Cysteine), (XX) C00133 (D-Alanine), (XXI) C00148 (L-Proline). Model simulation was performed from time 0 to 3600 seconds time period to examine temporal changes in metabolite concentrations using fourth-order Runge-Kutta method [37,38]. A simulation timestep of 0.1 seconds was employed and metabolite concentrations were constrained to the 0 millimolar to 1000 millimolar range throughout the simulation. Simulation outputs were sampled at 2 second intervals.

Results and Discussion

The resulting whole-cell KM of *B. breve* DSM 20213 developed using AdvanceSyn Model Specification as bbrdPS26; consists of 329 enzymes with 453 enzymatic reactions involving 580 metabolites [39]. This is consistent with the 327 enzyme-based whole-cell KM reported for *B. longum* [19]. In addition, 658 ODEs acting as placeholders for transcription and translation were added.

We deployed the bbrdPS26 model in AdvanceSyn Toolkit [36] and observed that it runs cleanly, as documented in Fig. 1. This demonstrates that bbrdPS26 is free of syntax and compilation errors as argued in recent model constructions [16,40-45]. Given the complexity typical of whole-cell kinetic frameworks, this indicates that the network architecture is internally consistent. The observation that L-glutamate is used to produce D-glutamate is likely a direct consequence of using uniform median values for enzyme kinetics rather than tailored or measured parameters [38]. Hence, this result should not be taken as a biological hypothesis even though a recent study suggests that KMs using median enzyme kinetics may also provide substantial predictability [46]. Nevertheless, the value lies in presenting a functional, testable whole-cell kinetic model of *B. breve* DSM 20213; which is ready for extension; for example, for more accurate kinetics, regulatory modules or analyses of metabolic resource allocation [47-51].

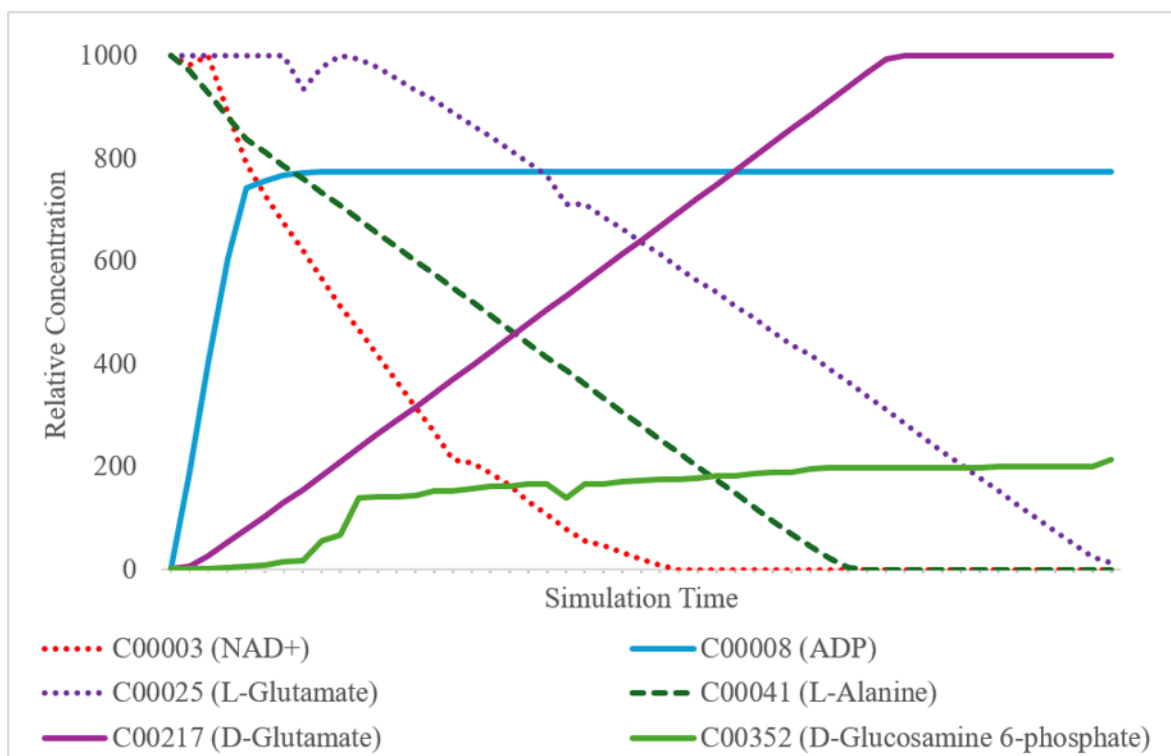


Figure 1: Simulation results of several metabolites against time.

Conclusion

The study presents the first ab initio whole cell kinetic model of *Bifidobacterium breve* DSM 20213 (bbrdPS26), comprising 580 metabolites, 329 enzymes with transcriptions and translations and 453 enzymatic reactions.

Supplementary Materials

Reaction descriptions and model can be downloaded from <https://bit.ly/bbrdPS26>

Conflict of Interest

The authors declared no potential conflicts of interest with respect to the research, authorship and/or publication of this article.

Funding Statement

This research did not receive any specific grant from funding agencies in the public, commercial or non-profit sectors.

Acknowledgement

The authors wish to thank the institute, Management Development Institute of Singapore, for its support towards this work. The cost of publication fees was borne by the authors.

Data Availability Statement

The data supporting the findings of this study are available from the corresponding author upon reasonable request.

Ethical Statement

The project did not meet the definition of human subject research under the preview of the IRB according to federal regulations and therefore was exempt.

Informed Consent Statement

Not applicable.

Authors' Contributions

All authors contributed equally to this paper.

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